EE5630 Digital Signal Processing

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Homework #6: FIR design by the window method

Due Monday 13 December 23:59

In this homework, you will design FIR lowpass filters. The filter specification is similar to HW5: cut-off at $\omega_c = 0.2\pi$, and edge of stopband at $\omega_s = 0.28\pi$.

1. Check the starter code, and write a function `myFIRrecwin(omega_c, N)` that returns the impulse response of a **type I lowpass filter** of length 2N+1 using the rectangular window (Type I means $M = 2N$ is an even number, and $w[M - n] = w[n]$).

2. Directly multiplying your result with any window other than the rectangular one would change the frequency response. Apply the Hann window and the Hamming window of length 2N+1, and compare the magnitude responses of your filters with the result given by the rectangular window. Summarize the differences.

3. Similar to Homework #5, let's set the stopband tolerance at $\delta = 0.01$, and $A = -20\log_{10}\delta$ = 40. In the starter code, fill in details for "`case 3`" to obtain the FIR impulse response that would satisfy the specs using a Kaiser window with appropriate β. How does the length of the filter compare to the case of using Hann or Hamming window?

4. Since the filter specification is similar to HW5, discuss whether it is more computationally efficient to apply IIR filtering (HW5) or FIR filtering (this homework). Discuss what other factors (delay, memory requirement, etc) might also affect the choice of IIR vs. FIR in practice.

Remarks: please recall that FIR can also be accelerated FFT in HW3.